Chapter 26: Production and Growth

“If two countries start with the same resources and technology today, what decisions made this decade could cause one to be ten times richer than the other in fifty years?”

Learning Objectives

26.1 Recognize how much economic growth differs around the world.

26.2 Identify why productivity is the key determinant of a country's standard of living and illustrate how to calculate it.

26.3 Define and apply the factors that determine a country's productivity.

26.4 List how a country's policies influence its productivity growth.

Key Points

- Economic prosperity, as measured by GDP per person, varies substantially around the world. The average income in the world's richest countries is more than ten times that in the poorest. Because growth rates of real GDP also vary substantially, the relative positions of countries can change dramatically over time.
- The standard of living depends on an economy's ability to produce goods and services. Productivity, in turn, depends on the amounts of physical capital, human capital, natural resources, and technological knowledge available to workers.
- Government policies can try to influence the economy's growth rate in many ways: by encouraging saving and investment, encouraging investment from abroad, fostering education, promoting good health, maintaining property rights and political stability, allowing free trade, and promoting the research and development of new technologies.

Key Points

- The accumulation of capital is subject to diminishing returns: The more capital an economy has, the less additional output it gets from an extra unit of capital. Although higher saving and investment leads to higher growth for a while, growth eventually slows down as capital, productivity, and income rise. Also because of diminishing returns, the return to capital is often high in poor countries. Other things equal, these countries can grow faster because of the catch-up effect.
- Population growth has various effects on economic growth. More rapid population growth may lower productivity by stretching the supply of natural resources and by reducing the amount of capital available for each worker. But a larger population may enhance the rate of technological progress because there are more scientists and engineers.

Incomes and Growth Around the World

FACT 1: There are vast differences in living standards around the world.

Country	RGDP/POP 1970	RGDP/POP 2017	Growth Rate
Botswana	$793	$16,235	1947.29%
Republic of Korea	$2,088	$34,955	1574.09%
Singapore	$5,797	$79,842	1277.30%
China	$1,415	$13,051	822.33%
Viet Nam	$785	$6,434	719.62%
India	$1,275	$6,281	392.63%
Chile	$7,110	$24,024	237.89%
Congo	$1,359	$4,581	237.09%
Germany	$14,905	$49,141	229.69%
Italy	$12,622	$39,470	212.71%
Japan	$12,912	$40,063	210.28%
Australia	$19,500	$48,141	146.88%
Greece	$10,390	$25,376	144.23%
United States	$23,473	$56,153	139.22%
Mexico	$8,093	$18,360	126.86%
Cambodia	$1,661	$3,485	109.81%
Jamaica	$6,575	$7,713	17.31%
Chad	$1,295	$1,368	5.64%
Venezuela (Bolivarian Republic of)	$8,681	$7,925	-8.71%
Zimbabwe	$2,469	$1,977	-19.93%

Incomes and Growth Around the World

FACT 2:

There is also great variation in growth rates across countries.

Country	RGDP/POP 1970	RGDP/POP 2017	Growth Rate
Botswana	$793	$16,235	1947.29%
Republic of Korea	$2,088	$34,955	1574.09%
Singapore	$5,797	$79,842	1277.30%
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Incomes and Growth Around the World

Since growth rates vary, the country rankings can change over time:

- Poor countries are not necessarily doomed to poverty forever, e.g. Singapore incomes were low in 1960 and are quite high now.
- Rich countries can't take their status for granted: They may be overtaken by poorer but faster-growing countries.
- **Rule of 70** – the approximate time it will take a country to double its GDP.
    - Found by dividing 70 by a country's growth rate

$$\#\,of\ Years = \frac{70}{Growth\ Rate}$$

Calculating Growth

Calculating growth:

Change in real GDP: $$\frac{RGDP_{\text{year 2}} - RGDP_{\text{year 1}}}{RGDP_{\text{year 1}}} \times 100\%$$

Ex (1) : RGDP in year 2 = \$50,000
RGDP in year 1 = \$45,000

$$\frac{\$50{,}000 - 45{,}000}{45{,}000} \times 100\%$$

$= 11.11\%$

Ex (2) : RGDP in year 2 = \$10,000
RGDP in year 1 = \$12,000

$$\frac{\$10{,}000 - 12{,}000}{12{,}000} \times 100\%$$

$= -16.67\%$

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Calculating Growth cont.

Calculating growth:

Change in real GDP per capita:

$$\frac{\frac{RGDP_{\text{year 2}}}{Pop_{\text{year 2}}} - \frac{RGDP_{\text{year 1}}}{Pop_{\text{year 1}}}}{\frac{RGDP_{\text{year 1}}}{Pop_{\text{year 1}}}} \times 100\%$$

Ex: RGDP in year 2 = \$50,000 Population in year 2 = 1,000
RGDP in year 1 = \$45,000 Population in year 1 = 950

$$\frac{\frac{\$50{,}000}{1{,}000} - \frac{45{,}000}{950}}{\frac{45{,}000}{950}} \times 100\% \quad = 5.6\%$$

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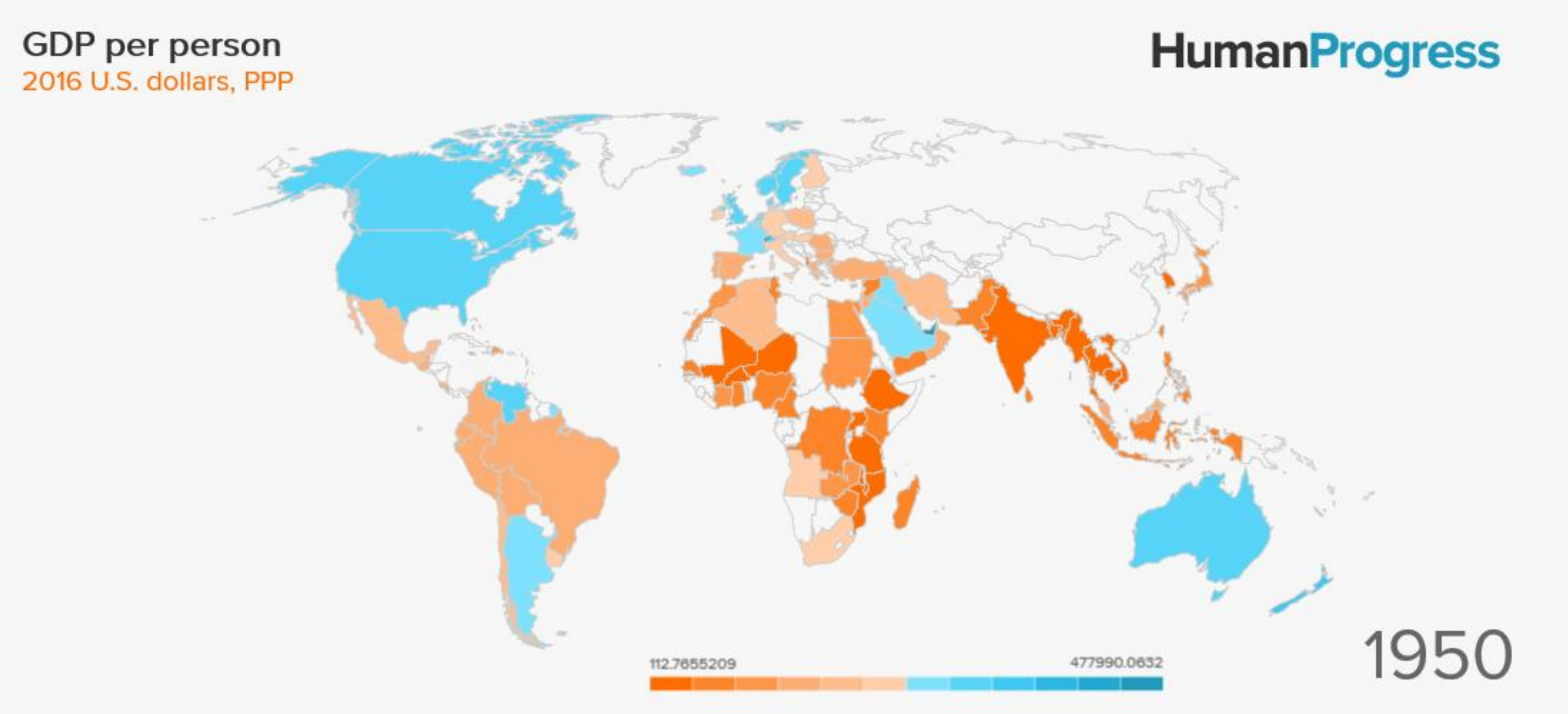
GDP per Capita 1950 v. 2017



Incomes and Growth Around the World

Questions:

- Why are some countries richer than others?
- Why do some countries grow quickly while others seem stuck in a poverty trap?
- What policies may help raise growth rates and long-run living standards?

Productivity

- Recall one of the Ten Principles from Chapter 1:

  A country's standard of living depends on its ability to produce goods and services.

- This ability depends on **productivity;** the average quantity of goods and services produced per unit of labor input.

- **Y** = real GDP = quantity of output produced
- **L** = quantity of labor
- so, productivity = **Y/L** (output per worker)

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Activity: Productivity

Pink Panther and Big Nose work 4,000 hours each and paint 800 poles total. Each pole is priced at $100

Q: What is their productivity?

Activity: Productivity

- Y / L = output per unit of labor or productivity
- Y = 800
- L = 4000 x 2 = 8000
- Y / L = 800 / 8000
- Y/ L = 0.10 poles per hour

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Activity: Productivity at the Dairy Farm

- Dairy Farms produces 1,080 gallons of milk per day.
- Each employee works 6 hours and has productivity of 20 gallons an hour.
- How many employees does Dairy Farms employ?

Activity: Productivity at the Dairy Farm

- Y = 1,080 gallons of milk per day
- L = 6 hours x # employees
- Y / L = 20 gallons per hour
- 1,080 gallons / 6 hours x # employees = 20 gallons per hour
- 1,080 gallons = 6h#e x 20 gallons per hour
- 1,080 gallons = 120 employee gallons
- 9 Employees

Why Productivity Is So Important?

- When a nation's workers are very productive, real GDP is large and incomes are high.
- When productivity grows rapidly, so do living standards.
- What, then, determines productivity and its growth rate?
  - Physical Capital
  - Human Capital
  - Natural Resources
  - Technology

Physical Capital Per Worker

- Recall: The stock of equipment and structures used to produce goods and services is called **[physical] capital**, denoted **K**.
- **K/L** = capital per worker.
- Productivity is higher when the average worker has more capital (machines, equipment, etc.).
- i.e., an increase in **K/L** causes an increase in **Y/L**.

Human Capital Per Worker

- **Human capital** (**H**):
  the knowledge and skills workers acquire through education, training, and experience

- **H/L** = the average worker's human capital

- Productivity is higher when the average worker has more human capital (education, skills, etc.).

- i.e., an increase in **H/L** causes an increase in **Y/L**.

Natural Resources Per Worker

- **Natural resources** (**N**): the inputs into production that nature provides, e.g., land, mineral deposits
- Other things equal,

  more **N** allows a country to produce more **Y**.

  In per-worker terms,
  an increase in **N/L** causes an increase in **Y/L**.
- Some countries are rich because they have abundant natural resources (e.g., Saudi Arabia has lots of oil).
- But countries need not have much **N** to be rich (e.g., Japan imports the **N** it needs).

Technological Knowledge

- **Technological knowledge**: society's understanding of the best ways to produce goods and services

- Technological progress does not only mean a faster computer, a higher-definition TV, or a smaller cell phone.

- It means any advance in knowledge that boosts productivity (allows society to get more output from its resources).
  - e.g., Henry Ford and the assembly line.

Technological Knowledge vs. Human Capital

- Technological knowledge refers to society's understanding of how to produce goods and services.

- Human capital results from the effort people expend to acquire this knowledge.

- Both are important for productivity.

The Production Function

- The production function is a graph or equation showing the relation between output and inputs:

$$\mathbf{Y} = \mathbf{A}\, \boldsymbol{F}(\mathbf{L}, \mathbf{K}, \mathbf{H}, \mathbf{N})$$

  ***F*( )** is a function that shows how inputs are combined to produce output

  "**A**" is the level of technology

- "**A**" multiplies the function **F**( ), so improvements in technology (increases in "**A**") allow more output (**Y**) to be produced from any given combination of inputs.

The Production Function

$$\mathbf{Y} = \mathbf{A}\,\boldsymbol{F}(\mathbf{L}, \mathbf{K}, \mathbf{H}, \mathbf{N})$$

- If we multiply each input by 1/**L**, then output is multiplied by 1/**L**:

$$\mathbf{Y/L} = \mathbf{A}\,\boldsymbol{F}(1, \mathbf{K/L}, \mathbf{H/L}, \mathbf{N/L})$$

- This equation shows that productivity (output per worker) depends on:
  - the level of technology (**A**)
  - physical capital per worker
  - human capital per worker
  - natural resources per worker

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Activity: The Engine of Growth

Using the simplified production function:

- $Y = A \times (\mathrm{K} \times H \times N)^{0.5} \times L^{0.5}$
- Calculate Country X's total output (Y).
- Suppose technology improves to **A = 2.5**. What happens to output?
- Which factor—capital, human capital, natural resources, or technology—seems to have the biggest potential for boosting Y?

Variable	Description	Quantity
L	Labor (workers)	100
K	Physical capital (machines)	50
H	Human capital (education index)	1.5
N	Natural resources index	1
A	Technological knowledge multiplier	2

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Activity: The Engine of Growth

1. $Y = 2 \times (50 \times 1.5 \times 1)^{0.5} \times (100)^{0.5}$
   $Y = 2 \times (75)^{0.5} \times 10 = 2 \times 8.66 \times 10 = 173.2$
2. With $A = 2.5$:
   $Y = 2.5 \times (75)^{0.5} \times 10 = 2.5 \times 8.66 \times 10 = 216.5$
3. Technology (A) amplifies all other factors—it multiplies the entire production function, showing why technological progress is critical for sustained growth.

If you were an economic advisor to Country X, where would you recommend investing—new machines, education, resource extraction, or R&D—and why?

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Economic Growth and Public Policy

Next, we look at the ways
public policy can affect
long-run growth in productivity
and living standards.

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Questions to Consider

Which of the following policies do you think would be most effective at boosting growth and living standards in a poor country over the long run?

a. Offer tax incentives for investment by local firms
b. ” ” ” ” ” by foreign firms
c. Give cash payments for good school attendance
d. Crack down on government corruption
e. Restrict imports to protect domestic industries
f. Allow free trade
g. Give away condoms

Saving and Investment

- We can boost productivity by increasing **K**, which requires investment.
- Since resources are scarce, producing more capital requires producing fewer consumption goods.
- Reducing consumption = increasing saving.
  This extra saving funds the production of investment goods.
  (More details in the chapter 26.)
- Hence, a tradeoff between current and future consumption.

Diminishing Returns and the Catch-Up Effect

- The government can implement policies that raise saving and investment. *(Details in chapter 26.)*

Then **K** will rise, causing productivity and living standards to rise.

- But this faster growth is temporary, due to

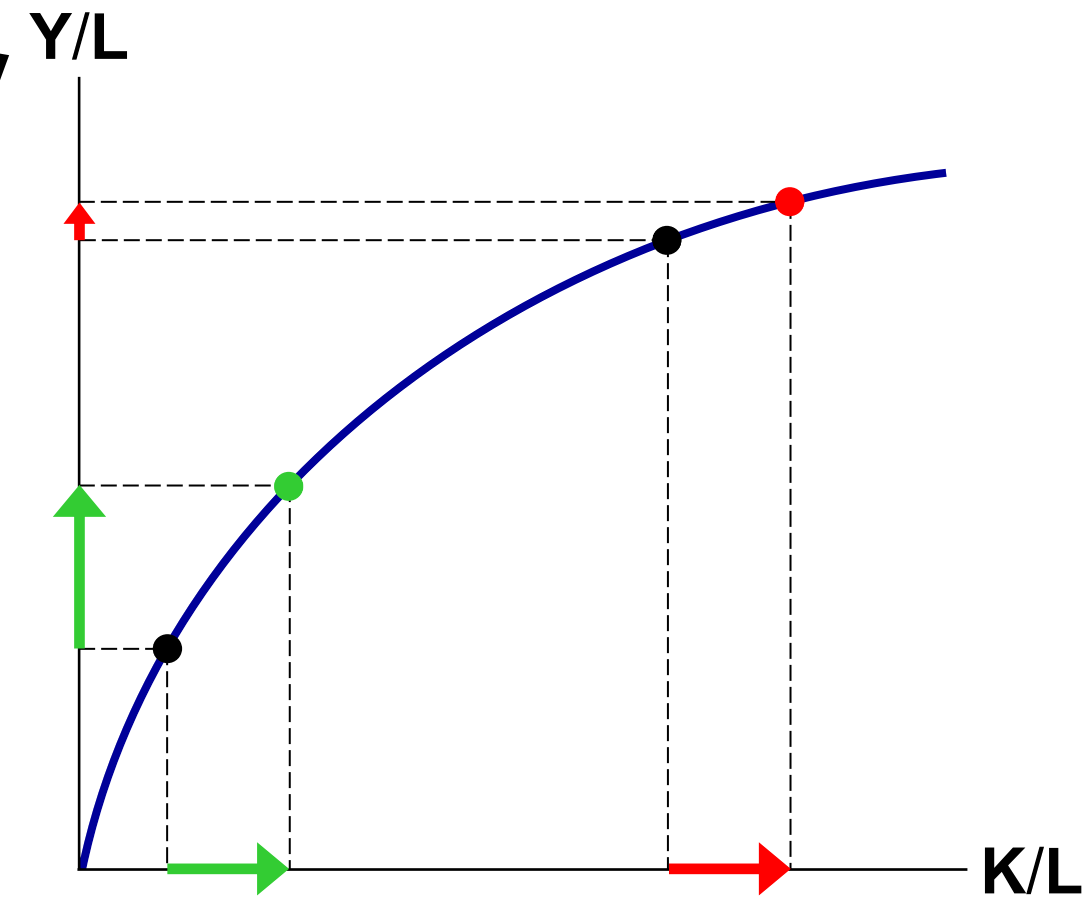
diminishing returns to capital:

As **K** rises, the extra output from an additional unit of **K** falls….

The Production Function & Diminishing Returns

If workers have little **K**, giving them more increases their productivity a lot.

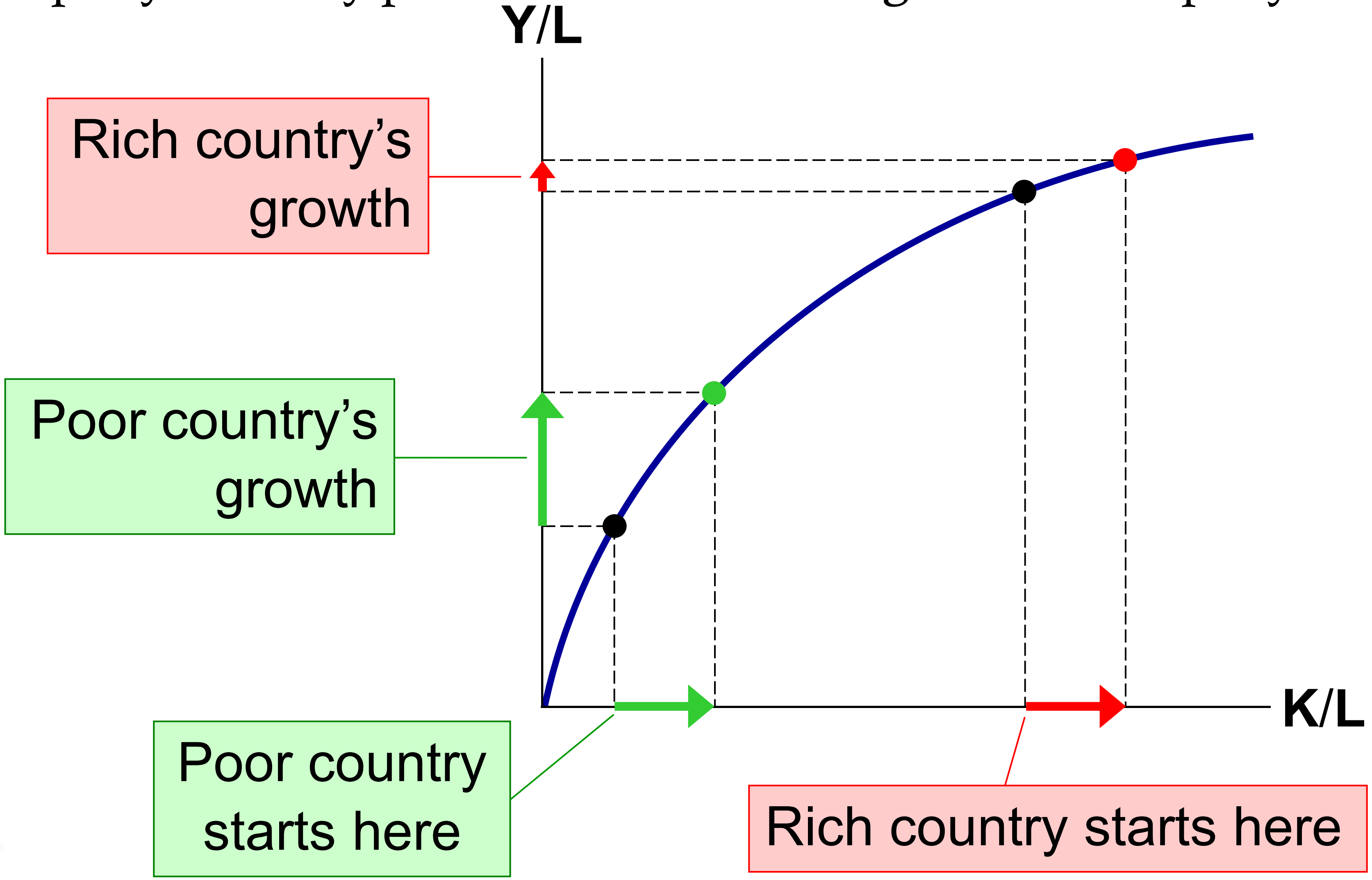
If workers already have a lot of **K**, giving them more increases productivity fairly little.



Capital per worker

The catch-up effect:

- the property whereby poor countries tend to grow more rapidly than rich ones



Activity: The Catch-Up Race

- Two nations, Alpha and Beta, both invest 20% of their GDP in new capital. Alpha's capital per worker is \$100,000; Beta's is \$20,000.
  - Which country should expect a higher growth rate in output per worker over the next decade?
  - Explain using the concept of diminishing returns.

Activity: The Catch-Up Race

- Beta
- Because its lower capital stock means each new unit of capital adds more to output (the catch-up effect).

The Catch-Up Effect in the Real World

- Over 1960–1990, the U.S. and S. Korea devoted a similar share of GDP to investment, so you might expect they would have similar growth performance.

- But growth was >6% in Korea and only 2% in the U.S.

- Explanation: the catch-up effect.
  In 1960, **K/L** was far smaller in Korea than in the U.S., hence Korea grew faster.

- China is another illustrative example.

Investment from Abroad

- To raise **K/L** and hence productivity, wages, and living standards, the government can also encourage
  - **Foreign Direct Investment**:
    a capital investment (e.g., a factory) that is owned & operated by a foreign entity
  - **Foreign Portfolio Investment**:
    a capital investment financed with foreign money but operated by domestic residents
- Some of the returns from these investments flow back to the foreign countries that supplied the funds.

Investment from Abroad

- Especially beneficial in poor countries that cannot generate enough saving to fund investment projects themselves.
- Also helps poor countries learn state-of-the-art technologies developed in other countries.

- OECD Study
- …a preponderance of studies shows that FDI triggers technology spillovers, assists human capital formation, contributes to international trade integration, helps create a more competitive business environment and enhances enterprise development. All of these contribute to higher economic growth, which is the most potent tool for alleviating poverty in developing countries.

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Education

- Government can increase productivity by promoting education–investment in human capital (**H**).
    - Public schools, subsidized loans for college
- Education has significant effects: In the U.S., each year of schooling raises a worker's wage by 10%.
- But investing in **H** also involves a tradeoff between the present & future:
    - Spending a year in school requires sacrificing a year's wages now to have higher wages later.
    - "Abel and Deitz also found that the return on a bachelor's degree, while averaging 15 percent for all college graduates, differs according to a student's major. In general, majors providing technical training earned the highest return. Engineering majors earned the highest return—21 percent—followed by math and computer majors and health majors (18 percent) and business majors (17 percent). At the other end of the scale, the return is 11 percent for leisure and hospitality majors and 9 percent for education majors."

Activity: Investing in Minds

- A country can either:
  - A) Subsidize college tuition, or
  - B) Build new factories.
- Which option likely raises long-run productivity more? Explain using human capital theory.

Activity: Investing in Minds

- A.
- Education raises workers' skills and knowledge, leading to sustained productivity growth. Factories help too, but their effect diminishes faster.

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Health and Nutrition

- Health care expenditure is a type of investment in human capital—healthier workers are more productive.
- In countries with significant malnourishment, raising workers' caloric intake raises productivity:
  - Over 1962–95, caloric consumption rose 44% in S. Korea, and economic growth was spectacular.
  - Nobel winner Robert Fogel: 30% of Great Britain's growth from 1790–1980 was due to improved nutrition.

Property Rights and Political Stability

- Recall: ***Markets are usually a good way to organize economic activity.***

- The price system allocates resources to their most efficient uses.

- This requires respect for **property rights**, the ability of people to exercise authority over the resources they own.

Property Rights and Political Stability

- In many poor countries, the justice system doesn't work very well:
  - Contracts aren't always enforced
  - Fraud, corruption often go unpunished
  - In some, firms must bribe government officials for permits
- Political instability (e.g., frequent coups) creates uncertainty over whether property rights will be protected in the future.

Property Rights and Political Stability

- When people fear their capital may be stolen by criminals or confiscated by a corrupt government, there is less investment, including from abroad, and the economy functions less efficiently.
- Result: lower living standards.

- Economic stability, efficiency, and healthy growth require law enforcement, effective courts, a stable constitution, and honest government officials.

Free Trade

- **Inward-oriented policies** (e.g., tariffs, limits on investment from abroad) aim to raise living standards by avoiding interaction with other countries.

- **Outward-oriented policies** (e.g., the elimination of restrictions on trade or foreign investment) promote integration with the world economy.

Free Trade

- Recall: ***Trade can make everyone better off.***
- Trade has similar effects as discovering new technologies—it improves productivity and living standards.
- Countries with inward-oriented policies have generally failed to create growth.
  - e.g., Argentina, North Korea during the 20th century.
- Countries with outward-oriented policies have often succeeded.
  - e.g., South Korea, Singapore, Taiwan after 1960.

Research and Development

- Technological progress is the main reason why living standards rise over the long run.

- One reason is that knowledge is a **public good**: Ideas can be shared freely, increasing the productivity of many.

- Policies to promote tech. progress:
  - Patent laws
  - Tax incentives or direct support for private sector R&D
  - Grants for basic research at universities

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Population Growth

…may affect living standards in 3 different ways:

1. Stretching natural resources

- 200 years ago, Malthus argued that pop. growth would strain society's ability to provide for itself.
- Since then, the world population has increased six-fold. If Malthus was right, living standards would have fallen. Instead, they've risen.
- Malthus failed to account for technological progress and productivity growth.

Population Growth

2. Diluting the capital stock

- Bigger population = higher **L** = lower **K/L** = lower productivity & living standards.
- This applies to **H** as well as **K**: fast pop. growth = more children = greater strain on educational system.
- Countries with fast pop. growth tend to have lower educational attainment.

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Population Growth (2 cont.)

2. Diluting the capital stock

To combat this, many developing countries use policy to control population growth.

- China's one child per family laws
- Contraception education & availability
- Promote female literacy to raise opportunity cost of having babies

Population Growth

3. Promoting tech. progress

- More people
  = more scientists, inventors, engineers
  = more frequent discoveries
  = faster tech. progress & economic growth
- Evidence from Michael Kremer:
  Over the course of human history,
  - growth rates increased as the world's population increased
  - more populated regions grew faster than less populated ones

Activity: People and Progress

- Suppose a nation’s population growth rate rises sharply. Identify one positive and one negative effect on long-run living standards, using concepts from this chapter.

Activity: People and Progress

- Positive: More people can spur innovation and technological progress.
- Negative: Resources and capital may be spread thinner, reducing productivity per worker.

Are Natural Resources a Limit to Growth?

- Some argue that population growth is depleting the Earth's non-renewable resources and thus will limit growth in living standards.
- But technological progress often yields ways to avoid these limits:
  - Hybrid cars use less gas.
  - Better insulation in homes reduces the energy required to heat or cool them.
- As a resource becomes scarcer, its market price rises, which increases the incentive to conserve it and develop alternatives.
- Peak oil debate

Effects on Growth Wrap-Up

Factor / Policy	Short-Run Effect on Growth	Long-Run Effect on Growth	Explanation
Saving & Investment	Boosts capital accumulation	Growth slows as diminishing returns set in	Higher K/L raises productivity initially
Foreign Investment	Increases capital stock quickly	Transfers technology, fosters innovation	Encourages knowledge spillovers
Education (Human Capital)	Temporarily lowers output as people study	Sustained increase in productivity	Improves worker skills and innovation
Health & Nutrition	Raises labor quality	Increases lifetime productivity	Healthy workers produce more
Property Rights & Stability	Encourages investment	Supports efficient resource allocation	Secure ownership promotes growth
Free Trade	Expands market access	Increases specialization and efficiency	Promotes technological diffusion
Research & Development	High upfront cost	Drives continuous productivity growth	Innovation expands "A" in production function
Population Growth	May dilute capital	Can boost tech progress if managed	Effect depends on balance between dilution and innovation

Key Takeaways

- Economic growth rates vary widely across countries, explaining global income differences.
- Productivity—output per worker—is the key determinant of a nation's standard of living.
- Productivity depends on four main factors: physical capital, human capital, natural resources, and technological knowledge.
- Policies that promote saving, investment, education, health, property rights, and open trade tend to accelerate long-run growth.
- Growth is limited by diminishing returns, but poor countries can grow faster through the catch-up effect.
- Population growth can both hinder (through resource dilution) and help (through innovation) long-run economic performance.

Knowledge Check

1. What is the primary determinant of a nation's standard of living?
2. List the four major factors of productivity.
3. Why does increasing physical capital yield diminishing returns?
4. How does the catch-up effect explain faster growth in poorer nations?
5. Give one example of how government policy can promote long-run growth.
6. How might rapid population growth hurt productivity?
7. How might it help growth in the long run?

Knowledge Check Answers

1. Productivity (output per worker).
2. Physical capital, human capital, natural resources, and technological knowledge.
3. Each additional unit of capital adds less to output as the existing stock grows.
4. Poorer countries grow faster because new capital has a larger marginal impact.
5. Examples: Encourage saving/investment, education, free trade, property rights, R&D.
6. It can dilute capital and strain natural resources.
7. It increases the number of potential innovators, boosting technological progress.